

Def. Define

$$f_a^c: [-1, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x^a \cdot \sin(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$

where $a \in \mathbb{R}, c > 0$. And, define quotient at $x=0$ of f_a^c as

$$\varphi_a^c: [-1, 1] \setminus \{0\} \rightarrow \mathbb{R}; t \mapsto \frac{f_a^c(t)}{t}$$

Then, $\varphi_a^c(t) = t^{a-1} \cdot \sin(|t|^{-c})$.

Prop. $\frac{d}{dx} f_a^c(x) = \begin{cases} \lim_{t \rightarrow 0} t^{a-1} \cdot \sin(|t|^{-c}) & \text{if } x=0, \text{ and the limit exists.} \\ a \cdot x^{a-1} \cdot \sin(|x|^{-c}) - c \cdot x^{a-1} \cdot |x|^{-c-2} \cdot \cos(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$

Simplify, $a x^{a-1} \cdot \sin\left(\frac{1}{x^c}\right) - c \cdot x^{a-c-1} \cdot \cos\left(\frac{1}{x^c}\right)$ if $x \neq 0$

$$2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

$$\lim_{t \rightarrow 0} t \cdot \sin$$

$$x^a \cos(|x|^{-c}) = -c \cdot x^{a-c-1}$$

$$+ x^a \cdot \cos(|x|^{-c}) \cdot \frac{x}{|x|^1} \cdot (-c) \cdot |x|^{-c-1}$$

Equivalent Condition Table	f	f'	f''
Continuous	$a > 0$	$a > 1+c$	$a > 2+2c$
bounded	$a \geq 0$	$a \geq 1+c$	$a \geq 2+2c$
Exists at $x=0$	Always...	$a > 1$	$a > 2+c$
Bounded Variation			

Prop.

Observe that $f: [-1, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x \sin(\frac{1}{x}) & \text{if } x \neq 0 \end{cases}$.

Then, f is Continuous, Not differentiable at $x=0$, Not bounded variation, Not bounded derivative.

Proof. Since $|x \sin(\frac{1}{x})| \leq |x| \rightarrow 0$ as $x \rightarrow 0$, clearly continuous.

Meanwhile, $f'(0)$ exists $\Leftrightarrow \lim_{t \rightarrow 0} \sin(\frac{1}{t})$ exists, thus not exists.

And, to show that f is Not bounded variation, Construct a Partition as:

$$p_n \stackrel{\text{def}}{=} \left\{ -1, 0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \dots, \frac{2}{2\pi}, \frac{2}{\pi}, 1 \right\}$$

Then, $V(f, p_n) \geq \sum_{k=0}^{n-1} \frac{2}{(2k+1)\pi} = \frac{2}{\pi} \cdot \sum_{k=0}^{n-1} \frac{1}{2k+1} \rightarrow \infty$ as $n \rightarrow \infty$.

Therefore it is Not bounded variation. Similarly, f_n^n ($n \in \mathbb{N}$) are Not bounded variation.

But, in this way, we can not determine $f_2'(x) = x^2 \sin(\frac{1}{x})$ is BV or Not.

Observe that

$$\frac{d}{dx} f_2'(x) = \begin{cases} 0 & \text{if } x=0 \\ 2x \sin(\frac{1}{x}) - \cos(\frac{1}{x}) & \text{if } x \neq 0 \end{cases}$$

It is Riemann integrable and bounded. Therefore, for any partition P on $[-1, 1]$,

$$V(f_2', P) = \sum_{i=1}^n |f_2'(x_{i-1}) - f_2'(x_i)| = \sum_{i=1}^n |f_2'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f_2'| \cdot \Delta x_i = \sup |f_2'| \cdot 2$$

$\uparrow$
MVT

Therefore, f_2' is Bounded Variation. Generally,

Prop. If $f: [a, b] \rightarrow \mathbb{R}$ is differentiable and the derivative f' is bounded, then f is bounded variation.

Proof. Ask a toddler on the street.

Prop. f_2^2 is not bounded variation.

Proof. Let $p_n = \{x_1, x_2, \dots, x_n, x_{n+1}\}$ as $x_1 = -1$, $x_{n+1} = 1$, and

$$x_k = \frac{1}{\sqrt{(\frac{1}{2} + n - k)\pi}} \quad 0 \leq k \leq n$$

$$\frac{1}{x^2} = \left(\frac{1}{2} + n\right)\pi$$

$$x = \frac{1}{\sqrt{(\frac{1}{2} + n)\pi}}$$

Then, $V(f_2^2, p_n) \geq \sum_{i=1}^n |f_2^2(x_i) - f_2^2(x_{i-1})|$

$$= \sum_{i=1}^n \frac{1}{\left(\frac{1}{2} + n - i\right)\pi} + \frac{1}{\left(\frac{1}{2} + n - (i-1)\right)\pi}$$

$$\geq 2 \cdot \sum_{i=1}^n \frac{1}{\left(\frac{1}{2} + n - i + 1\right)\pi}$$

$$= \frac{2}{\pi} \cdot \sum_{i=1}^n \frac{1}{\frac{1}{2} + i} \geq \frac{2}{\pi} \sum_{i=1}^n \frac{1}{i} \rightarrow \infty \quad \text{as } n \rightarrow \infty$$